Dihedral centralizers and regular polytopes over a projective line

Private author manuscript, phase 230. This is a rederivation of two specified principal source contracts, not an independent priority claim for their conclusions. Independent mathematical review is pending.

1. The results and the common argument

Let q be a prime power. We write $L(q) = \mathrm{PSL}_2(q)$ and $P(q) = \mathrm{PGL}_2(q)$. A string C-group is a group $G = \langle r_0, \dots, r_{n-1} \rangle$ with distinct involutory generators, with $r_i r_j = r_j r_i$ when $|i - j| > 1$, and with the intersection property

$$G_I \cap G_J = G_{I \cap J}, \qquad G_I = \langle r_i : i \in I \rangle. \tag{1}$$

The empty subgroup is trivial. In particular, no generator belongs to the subgroup generated by the others. The type of a string C-group is the sequence of orders of consecutive generator products.

We prove the following classification, originally established by Leemans and Schulte in the two sources specified in §9.

Theorem 1. The regular abstract polytopes of rank at least four whose full automorphism groups are projective special or general linear groups of dimension two are precisely:

Full group	Rank	Type	Facets	Vertex figures
$L(11)$	4	$\{3,5,3\}$	hemi-icosahedra	hemi-dodecahedra
$L(19)$	4	$\{5,3,5\}$	hemi-dodecahedra	hemi-icosahedra
$P(5) \cong S_5$	4	$\{3,3,3\}$	tetrahedra	tetrahedra

There is one isomorphism class in each row. The first two are the 11-cell and the 57-cell; the last is the 4-simplex. Each is self-dual. There are no examples of rank at least five and no further examples in even characteristic.

The proof separates into a uniform reduction and a finite calculation. The reduction uses only elementary two-dimensional linear algebra and the *types* in Dickson's subgroup classification. We do not require the numbers of conjugacy classes of subfield groups or the incidence counts of dihedral groups inside them. The finite calculation concerns six rank-three markings and ten amalgams, of order at most 3420. Its exact presentations and a runnable GAP calculation are included.

The argument that replaces the subfield incidence counts is the following observation. A nonabelian dihedral subgroup defined over a field k has no new centralizing projectivities after extending the field: its centralizer is trivial in the unipotent case and is generated by a single projectivity already defined over k in the semisimple case. In a string marking, the next generator must lie in this centralizer. Consequently, a subfield facet is normalized by that generator. An odd adjacent product then forces the generator back into the facet, contradicting (1).

This is a common proof of the two classifications. It does not reprove the classical subgroup classification, and it makes no claim to have settled a new open conjecture. The formal and computational boundaries are stated in §8.

2. Retained inputs and elementary string consequences

We use two established background theorems.

Input D (Dickson's classification, in its projective form). For a finite field K of odd characteristic, a proper subgroup of $L(K)$ or $P(K)$ is one of the following types: cyclic; dihedral; elementary abelian of characteristic order; a subgroup of an affine group $V \rtimes k^\times$; A_4, S_4, or A_5; or a conjugate of $L(k)$ or $P(k)$ for a subfield $k \subseteq K$. Only the types that actually embed are intended. The subfield alternatives are conjugate to the usual matrix embeddings, not merely abstractly isomorphic groups. We also retain the usual orders

$$|L(q)| = \frac{q(q^2-1)}{(2, q-1)}, \qquad |P(q)| = q(q^2-1), \tag{2}$$

the simplicity of $L(q)$ for $q \geq 4$, and the trivial centralizer of $L(q)$ in $P(q)$.

The required subgroup list is printed as Theorem 2 in L083 and as Propositions 1–2 in L079, with attribution there to Dickson, Moore and the later projective subgroup account. Their subgroup multiplicity formulas are not inputs to our proof. In particular the affine alternative includes dihedral groups of order $2p$, where $p = \operatorname{char} K$; it must not be silently replaced by the semisimple torus cases.

Input P (the string C-group correspondence). String C-markings, up to group isomorphisms preserving the ordered marking, correspond to regular abstract polytopes. The faces of rank i are the cosets of the subgroup generated by all generators except r_i. Reversing the marking gives the dual polytope. This is the standard correspondence used in both sources and in the existing proof library.

We first dispose of a possible degeneracy. If a consecutive product has order two, all generators on one side of that place commute with all generators on the other. By (1) the two generated subgroups meet trivially. Since they generate G, the multiplication map from their direct product to G is an isomorphism. Thus a directly indecomposable group has only types with consecutive entries at least three.

All the groups $L(q)$ and $P(q)$ are directly indecomposable. For $q \geq 4$, simplicity proves this for $L(q)$. If $P(q) = A \times B$, its simple normal subgroup $L(q)$ must lie in one factor: its projections have commuting normal images, and a simple nonabelian group cannot have two nontrivial such images. More explicitly, $L(q) \cap A$ and $L(q) \cap B$ are normal, and if both were trivial, the commutators of $L(q)$ with both factors would be trivial. The other factor then centralizes $L(q)$, so is trivial. In even characteristic $P(q) = L(q)$. The remaining groups are S_3, A_4, S_4, for which the elementary normal-subgroup lists give the same conclusion. These lists are $1, A_3, S_3$; $1, V_4, A_4$; and $1, V_4, A_4, S_4$, respectively.

Every marking of rank at least four in these groups therefore has all consecutive product orders at least three. Restricting a marking to an interval of three indices gives an irreducible rank-three string C-group.

Lemma 2 (normalizing a facet). Suppose $G = \langle H, s\rangle$, $s^2 = 1$, and $sHs^{-1} = H$. If $t \in H$ and ts has odd order, then $s \in H$.

Proof. The normalizer assumption makes $H \lhd G$. The quotient is generated by sH, whose order divides two. Since $tH = H$, the order of the same element $tsH = sH$ divides the odd order of ts. Hence $sH = H$. $\square$

This lemma is valid without a finiteness assumption on G, provided the displayed odd order exists. It is the common final step in both subfield exclusions.

3. Centralizers on the projective line

We prove the matrix facts used in the reduction. All projectivities act on the projective line; equality of matrices representing them is understood up to nonzero scalar.

Lemma 3 (characteristic two). In $P(K)$, for K of characteristic two, the centralizer of every nonidentity involution is abelian.

Proof. Over an algebraic closure an involution is unipotent, since a semisimple element of order two would have eigenvalue ratio $-1 = 1$. Conjugate it to $u : x \mapsto x+1$. It has the unique fixed point infinity, so a commuting projectivity must fix infinity and has the form $x \mapsto ax + b$. Commutation gives $a = 1$. These translations commute with one another. The centralizer over K is a subgroup of this abelian group. □

Lemma 4 (commuting involutions in odd characteristic). Let K have odd characteristic. The centralizer of a four-group in $P(K)$ is that four-group. In particular $P(K)$ contains no elementary abelian subgroup of order eight.

Proof. We may extend scalars to an algebraic closure. Put the first involution in the form $z : x \mapsto -x$. A projectivity commuting with z either preserves or interchanges its two fixed points, and accordingly has the form $x \mapsto ax$ or $x \mapsto a/x$. The only nontrivial involution of the first form is z. An independent commuting involution is therefore of the second form; rescaling the coordinate makes it $w : x \mapsto 1/x$. The projectivities commuting with both z and w are exactly

$$x, \quad -x, \quad 1/x, \quad -1/x.$$

These are the given four-group. Since its elements were already defined over K, intersecting with $P(K)$ changes nothing. □

Here is the field-extension statement responsible for the common reduction.

Theorem 5 (dihedral centralizer descent). Let $k \subseteq K$ be fields of odd characteristic. Let $a, b \in P(k)$ be distinct involutions such that $u = ab$ has finite order greater than two. Then

$$C_{P(K)}(\langle a, b\rangle) \ \subseteq \ P(k). \tag{3}$$

If u is unipotent, this centralizer is trivial. Otherwise it has order two. In the semisimple case, if $A \in \mathrm{GL}_2(k)$ represents u, its nontrivial element is represented by

$$Z = A - \tfrac{1}{2}\operatorname{tr}(A)I. \tag{4}$$

Proof. First suppose u is semisimple. Over an algebraic closure it has two distinct eigenlines, and we may write it as $x \mapsto \lambda x$, where $\lambda \neq \pm 1$. A projectivity commuting with u must preserve its individual fixed points: an interchange would conjugate u to $u^{-1} \neq u$. Its form is therefore $x \mapsto cx$. Since $aua^{-1} = u^{-1}$, the involution a interchanges the eigenlines and has the form $x \mapsto d/x$. Commutation of $x \mapsto cx$ with a gives $c^2 = 1$. Thus the centralizer over the algebraic closure is precisely $\{1, z\}$, where $z : x \mapsto -x$.

Write the eigenvalues of A as α, β. The matrix in (4) has eigenvalues $(\alpha-\beta)/2$ and $(\beta-\alpha)/2$. Both are nonzero, and their ratio is -1. Thus (4) represents z, is invertible, and has entries in k. This

proves both the exact order and (3), including the nonsplit torus case. The expression is unaffected projectively if A is rescaled.

Now suppose u is unipotent. Over the algebraic closure put $u : x \mapsto x + 1$. As in Lemma 3 its centralizer consists of translations $x \mapsto x + c$. An involution conjugating u to u^{-1} has form $x \mapsto -x + d$. A translation commuting with it satisfies $2c = 0$, hence $c = 0$. The centralizer is trivial. These two cases exhaust nonidentity elements of $P(k)$ in positive characteristic. □

The unipotent case explains directly why a dihedral group of order $2p$ cannot have an independent commuting involution in odd characteristic. This is the case singled out in the corrective remark following Proposition 2 of L079. No absence of dihedral groups of order $2p$ is assumed here.

Corollary 6 (odd end labels). In a rank-four string C-group contained in $P(K)$, with K of odd characteristic and all consecutive labels at least three, the orders of r_0r_1 and r_2r_3 are odd.

Proof. Put $D = \langle r_0, r_1 \rangle$ and let $t = |r_0r_1|$. The generator r_3 centralizes D and lies outside it by (1). Theorem 5 first excludes the unipotent case. If t were even, the nontrivial involution $(r_0r_1)^{t/2} \in D$ would centralize D. The centralizer has exactly one nonidentity element, so it would equal r_3, a contradiction. Reverse the indices for the other end label. □

4. The uniform rank and facet reduction

Proposition 7 (rank restriction). A string C-group isomorphic to $L(q)$ or $P(q)$ has rank at most four. If its rank is four, q is odd.

Proof. In odd characteristic a marking of rank at least five contains the independent commuting involutions r_0, r_2, r_4. Their generated subgroup is elementary abelian of order eight: (1) excludes every dependence. This contradicts Lemma 4.

In characteristic two, a marking of rank at least four has $r_2, r_3 \in C_G(r_0)$. Lemma 3 implies that they commute, contrary to the consecutive label being at least three. This also excludes all higher ranks. □

Lemma 8 (the nonexceptional rank-three groups). An irreducible rank-three string C-group cannot be cyclic, elementary abelian, dihedral, A_4, or a subgroup of the one-dimensional affine group in odd characteristic.

Proof. Cyclic and elementary abelian groups are immediate. The involutions of A_4 generate its four-group and cannot generate A_4.

In an affine group in odd characteristic, an involution must have multiplier -1, so has form $x \mapsto -x + v$. Two such involutions commute only when their translation parameters agree. Thus there cannot be two distinct commuting end generators.

For a dihedral group use its usual rotation/reflection description. A central end generator would give an adjacent label two. Otherwise both end generators are reflections. Two distinct commuting reflections have product the central involution z, so $r_2 = zr_0$. The middle generator cannot be central, again by irreducibility, and hence is a reflection. Put $u = r_0r_1$. Then

$$(r_1r_2)^2 = (u^{-1}z)^2 = u^{-2}.$$

The nonidentity rotation u^2, nonidentity because $|u| > 2$, belongs to both adjacent rank-two subgroups. It does not belong to the reflection subgroup $\langle r_1 \rangle$. This violates (1). □

Theorem 9 (exceptional facets). Let $G \cong L(q)$ or $P(q)$ have a rank-four string C-marking. Its facet and vertex-figure groups are isomorphic to S_4 or A_5, and are proper subgroups.

Proof. Proposition 7 gives odd characteristic. Let $H = \langle r_0, r_1, r_2 \rangle$. It is proper by (1) and irreducible by §2. Input D and Lemma 8 leave only S_4, A_5, and the conjugates of standard subfield groups $L(k)$ and $P(k)$.

Suppose H is a subfield group, and conjugate the embedding so that it is standard. The two involutions $r_0, r_1 \in H \subseteq P(k)$ have product order greater than two. Theorem 5 places their centralizer in $P(k)$. Since r_3 centralizes them, $r_3 \in P(k)$. If $H = P(k)$, this immediately contradicts (1). If $H = L(k)$, the determinant-square homomorphism shows that $P(k)$ normalizes H. Hence r_3 normalizes H. Corollary 6 makes $|r_2 r_3|$ odd, so Lemma 2, with $t = r_2$, again gives $r_3 \in H$, a contradiction. Reversing the marking proves the vertex-figure assertion. □

No comparison of the extension degrees is needed. This proof includes subfield embeddings of degree two, arbitrarily large extension degrees, and the small subfield groups whenever the subgroup list permits them. Exceptional groups abstractly isomorphic to $P(3)$ or $L(5)$ in a *different* characteristic are not treated as subfield embeddings; they correctly remain among S_4, A_5.

5. The exceptional finite calculation

For integers m, n, k, write $[m, n]_k$ for the presentation

$$\langle x, y, z \mid x^2 = y^2 = z^2 = (xz)^2 = (xy)^m = (yz)^n = (xyz)^k = 1 \rangle. \tag{5}$$

The corresponding map, when it is a string C-group, is traditionally written $\{m, n\}_k$. This is the Petrie relation convention of L083, physical page 9.

Finite Lemma 10. Up to automorphisms preserving the ordered marking, the irreducible rank-three string C-markings of S_4 and A_5 have precisely the following triples (m, n, k):

$$\begin{array}{c|ccc} S_4 & (3,3,4) & (3,4,3) & (4,3,3) \\ A_5 & (3,5,5) & (5,3,5) & (5,5,3). \end{array} \tag{6}$$

In each case (5) is a presentation of the indicated group.

Finite proof and audit procedure. The program enumerates every involution of S_4 and A_5, every ordered pair of distinct commuting involutions for x, z, and every middle involution y. It tests generation, both adjacent orders, and (1) on all eight subsets. There are 72 surviving ordered triples in S_4 and 360 in A_5. It partitions these by the full automorphism group, not merely inner conjugation, and obtains the three ordered types in each row of (6). It separately constructs (5) as a finitely presented group and checks order 24 or 60. A marking gives a surjection from (5) onto the corresponding finite group; equality of orders makes it an isomorphism. These are exhaustive finite calculations with GAP's exact subgroup and finite-presentation machinery. □

For two entries $f = (m, n, k)$ and $v = (n, s, l)$ from (6), define

$$\begin{aligned} U(f, v) = \langle r_0, r_1, r_2, r_3 \mid & r_i^2 = 1,\ (r_i r_j)^2 = 1\ (j - i > 1), \\ & (r_0 r_1)^m = (r_1 r_2)^n = (r_2 r_3)^s = 1, \\ & (r_0 r_1 r_2)^k = (r_1 r_2 r_3)^l = 1 \rangle. \end{aligned} \tag{7}$$

The marking of every rank-four polytope with these local maps is a quotient of (7). Matching the common middle label in (6) gives fourteen ordered pairs. Reversal sends (f, v) to (v^*, f^*), where

$(a,b,c)^* = (b,a,c)$. There are ten orbits, including six fixed pairs; the following table contains one from each orbit.

Row	Facet f	Vertex figure v	$\lvert U(f,v)\rvert$	Group	String C-marking
1	$(5,5,3)$	$(5,5,3)$	1	trivial	no
2	$(5,5,3)$	$(5,3,5)$	1	trivial	no
3	$(5,3,5)$	$(3,5,5)$	3420	$L(19)$	yes
4	$(5,3,5)$	$(3,4,3)$	60	A_5	no
5	$(5,3,5)$	$(3,3,4)$	1	trivial	no
6	$(4,3,3)$	$(3,4,3)$	96	$2^4 : S_3$	yes
7	$(4,3,3)$	$(3,3,4)$	24	S_4	no
8	$(3,5,5)$	$(5,3,5)$	660	$L(11)$	yes
9	$(3,4,3)$	$(4,3,3)$	1	trivial	no
10	$(3,3,4)$	$(3,3,4)$	120	S_5	yes

Finite Lemma 11. The table is correct. The groups in rows 3, 8 and 10 admit automorphisms reversing their four specified generators. The face counts in these rows, from vertices through facets, are respectively

$$(57,171,171,57), \quad (11,55,55,11), \quad (5,10,10,5). \tag{8}$$

The graph of vertices and edges in row 8 is the complete graph K_{11}.

Finite proof and audit procedure. For each literal presentation (7), the program computes the order of the finitely presented group, constructs an isomorphic permutation group, and transports each specified generator through that isomorphism. It tests (1) on all sixteen subsets, including nonconsecutive ones. Thus a group identification alone does not certify a polytope. The identity group is handled separately, since GAP returns `fail` for its generic finitely presented permutation-isomorphism request.

The three main rows are identified by an actual group isomorphism with $L(19), L(11), S_5$, respectively. Reversal is checked by constructing the homomorphism on the four generators and checking bijectivity. The program computes the four maximal-parabolic indices, giving (8). For row 8 it constructs the actual vertex and edge cosets and their incidences. Every edge has two endpoints and the resulting 55 unordered pairs are all distinct, proving the K_{11} assertion.

For the auxiliary row 6 it verifies an elementary abelian normal subgroup of order 16, quotient S_3, and the existence of a complement. The indicated marking has face counts $(4,6,6,4)$. The failures in rows 4 and 7 are checked by the intersection test; independently, a stipulated facet or vertex-figure group already has the order of the full group, so it cannot occur as the required proper parabolic. The remaining failures are trivial groups. □

This calculation recovers the actual ten-row table of L083, rather than selecting only its two successful linear-group rows. The six local maps and ten amalgams are prior results; the program supplies a new replay of them.

6. Proof of the two classifications

We now complete Theorem 1, including the quotient issue. The fact that a group occurs as a universal group in the table would not on its own exclude other admissible quotients.

Suppose first that $G = L(q)$ is a rank-four string C-group. By Theorem 9 it is a quotient of one of the ten groups in the table. The trivial rows give nothing. Rows 4 and 7 cannot yield a rank-four marking with the stipulated facet, since an end parabolic already fills the universal group; passing to a quotient cannot restore the missing independent generator.

In row 6, the order of G divides 96. By (2), no $L(q)$ with $q \geq 4$ has order dividing 96. The small cases $L(2) = S_3, L(3) = A_4$ cannot contain a facet S_4 or A_5. In row 10 the proper nontrivial quotient of S_5 is C_2; this follows from the elementary normal-subgroup list $1, A_5, S_5$. Thus no projective special linear case comes from that row.

The groups in rows 3 and 8 are simple, so a nontrivial quotient is the whole group. Formula (2), which is strictly increasing within the odd prime powers, gives $q = 19$ and $q = 11$, respectively. Their markings pass the full intersection test. This proves existence and exhaustion for $L(q)$.

For $G = P(q)$, even characteristic has already been excluded. The same facet argument excludes rows 4 and 7. A nontrivial quotient of rows 3 or 8 is simple of the displayed order and cannot be $P(q)$, whose determinant-square quotient has order two for odd q. In row 6, (2) permits only $q = 3$: the next odd prime power, 5, already gives order 120, and $24 = |P(3)|$ divides 96. But $P(3) \cong S_4$ cannot have its required S_4 facet as a proper subgroup. In row 10 the only nontrivial quotients are S_5, C_2, so $q = 5$ and $G = P(5) \cong S_5$.

Uniqueness also follows at the level of markings. Every local marking is one of (6), and each admissible pair has the exact universal presentation (7). In the simple cases every nontrivial quotient map is an isomorphism. In the S_5 case the proper quotient is too small. Thus each surviving row determines one marked-group isomorphism class. The reversal automorphisms identify its dual with itself. Input P then proves Theorem 1. □

For clarity, the very small groups require no unstated use of simplicity: in a putative rank-four marking their proper facets would have to be S_4 or A_5, which immediately excludes S_3, A_4, S_4 as full groups.

7. The geometric conclusions and the scope of universality

The presentations for the two locally projective examples can be written uniformly as the string Coxeter relations of type $\{3, 5, 3\}$ or $\{5, 3, 5\}$, together with

$$(r_0 r_1 r_2)^5 = (r_1 r_2 r_3)^5 = 1. \tag{9}$$

The local presentations in (6) identify the facet and vertex-figure maps as the hemi-icosahedron and hemi-dodecahedron in the indicated order. These standard names refer to the antipodal quotients of the ordinary icosahedron and dodecahedron. Their interpretation is the same as in L083, §5; no new geometric construction is being attributed to this paper.

Since the universal group (7) itself satisfies (1), Input P supplies a regular polytope with exactly those local maps. Every other regular polytope with those local maps has a marking quotient of (7). The canonical marked-group epimorphism induces the corresponding polytope covering. Thus these two examples are universal with the specified facets and vertex figures. This is a statement about these *polytopal* universal groups, not a claim that every abstract facet amalgam is automatically polytopal.

The first two face vectors in (8) give the stated eleven or fifty-seven facets and vertices. The 11-cell has 55 edges and triangular two-faces; the 57-cell has 171 edges and pentagonal two-faces. The local

projective-plane maps justify the usual description as locally projective. The full groups coincide with their even-generator subgroups, because a subgroup of index at most two in a nonabelian simple group cannot have index two.

For row 10 the adjacent transpositions $(12), (23), (34), (45)$ give the standard simplex marking of S_5. This identifies the unique resulting polytope with the 4-simplex, independently of its face numbers alone.

The prior constructions of the 11-cell by Grünbaum and of the 57-cell by Coxeter remain the historical sources of these objects. Their construction history is not a novelty claim here.

8. Verification and what it does not establish

The written uniform argument is given in §§2–4. Its retained substantive input is Dickson's subgroup classification, plus the standard string C-group correspondence and the elementary classical group facts explicitly listed in §2.

The finite part is `phase230_finite_check.g`. Its exhaustive loops enumerate actual involutions and actual subgroup intersections. The finite-presentation order computations rely on GAP's exact algorithms. They are stronger than checking relations in one chosen permutation image, since such an image would give only an order lower bound. They are not yet a kernel-checked Lean replay of the ten universal presentations. The certificate interface must therefore continue to name GAP as a trusted computational dependency.

The separate Lean component `phase230_formal/DihedralFacetObstruction.lean` verifies five abstract declarations: an involution with an odd vanishing power is trivial; the corresponding homomorphism statement; membership in a normal facet; derivation of that membership from generation and the normalizer condition; and the contradiction with an independent extra generator. It does not verify the two-dimensional centralizer calculation, Dickson's theorem, all of Input P, or the finite presentation enumerations. No fully formal proof of Theorem 1 is claimed at this author checkpoint.

The portable replay alters a facet Petrie relation and rejects the resulting wrong universal order. A second control leaves the computed group intact but duplicates a distinguished generator; the marking test rejects it. The source presentations, expected orders and subgroup tests are exposed so an independent reviewer can inspect the mathematical bridge directly.

9. Source contracts, attribution and reuse

The two sources have been read in full in their retained primary author editions:

- **L083:** D. Leemans and E. Schulte, *Groups of type $L_2(q)$ acting on polytopes*, retained arXiv `math/0606660v1`, 14 physical pages. The principal numbered claims are Theorems 1, 3 and 4. Physical pages 2 and 9–10 state the rank-four classification, uniqueness and geometric identification; page 2 also records the exclusion of rank at least five. The current proof recovers those claims, the ten-row Table 1, and the geometric properties just derived.
- **L079:** D. Leemans and E. Schulte, *Polytopes with groups of type $PGL_2(q)$*, retained arXiv `0909.1991v1`, 9 physical pages. The principal classification is Theorem 10 on physical page 7, including the rank restriction of Lemma 3 and the introductory rank-at-least-four formulation. The current proof recovers that contract.

The source subgroup tables remain retained classical inputs. We do not claim to have rederived every auxiliary lemma in either source: the subfield-intersection lemmas are replaced by Theorem 5 and Lemma 2, so they are unnecessary premises. The general centralizer strategy is consistent with the earlier Suzuki proof bank, specifically phase164 §4.3; the projective-line calculation and the single facet-normalizer reduction are newly written here relative to that bank. They should be described as a reorganized proof, without an unsupported claim of historical novelty.

The six exceptional local maps, the amalgam table, the classification conclusions, and the names and properties of the two locally projective examples belong to the earlier literature. Their new contribution here is a common derivation with an inspectable exact computation. Classical material is retained as named input; the present proof supplies the matrix descent, the uniform assembly and the certificate interface.

The following printed material has different status and is **not silently included** in these two principal contracts:

1. L083, §5, describes the larger one-Petrie-relation group $J_1 \times L(19)$, referring to Hartley–Leemans, and the one-relation $L(11)$ phenomenon, referring to Grünbaum/Hartley. Those are background results from other sources, not proved here.
2. L083, §6, and L079's final discussion recall chiral or two-flag-orbit constructions obtained by modular reduction. They are attributed there to earlier authors. The present classification concerns *full flag-transitive* automorphism groups and does not rederive those constructions.
3. L079, physical pages 7–8, prints a computer census of seventeen rank-four polytopes with group $P\Sigma L_2(25)$, together with their types. This is a genuine additional numerical assertion outside the two linear-group classifications. It is not presently reproduced by this finite program and must remain an unpaid secondary source clause.
4. L079 ends with Conjecture 11 on other almost-simple extensions for odd square q. Neither part is an open problem solved by this manuscript; its subsequent status requires a separate primary-literature audit.

Consequently this author package proposes two *principal classification* contracts for independent review. It proposes no whole-paper compression ratio, no automatic full-source status, and no new-conjecture count. Any length comparison must include the retained theorem conventions and the finite computational component actually needed by the proof.